PRODUCT USER MANUAL

For OCEANCOLOUR_HR Products

OCEANCOLOUR_ARC_BGC_HR_L3_NRT_009_201
OCEANCOLOUR_BAL_BGC_HR_L3_NRT_009_202
OCEANCOLOUR_NWS_BGC_HR_L3_NRT_009_203
OCEANCOLOUR_IBI_BGC_HR_L3_NRT_009_204
OCEANCOLOUR_MED_BGC_HR_L3_NRT_009_205
OCEANCOLOUR_BLK_BGC_HR_L3_NRT_009_206
OCEANCOLOUR_ARC_BGC_HR_L4_NRT_009_207
OCEANCOLOUR_BAL_BGC_HR_L4_NRT_009_208
OCEANCOLOUR_NWS_BGC_HR_L4_NRT_009_209
OCEANCOLOUR_IBI_BGC_HR_L4_NRT_009_210
OCEANCOLOUR_MED_BGC_HR_L4_NRT_009_211
OCEANCOLOUR_BLK_BGC_HR_L4_NRT_009_212

Issue: 3.0

Contributors: K. Stelzer, D. Van der Zande, M. Böttcher, C. Lebreton

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MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2021/01/15	all	Initial version EIS May 2021	K. Stelzer, D. Van der Zande, M. Böttcher	V.Brando, E. Böhm
1.1	2021/04/07	all	Removal of MYNRT for compliance with PIT and QuID. Delivery mechanism changed from “Subsetter” to “FTP”.	K. Stelzer, D. Van der Zande, M. Böttcher	E.Böhm; V. Brando
1.2	2021/06/24		Inclusion of description of full-region mosaic datasets	K. Stelzer	
1.3	2022/03/30		upgrade of processing baseline for Sentinel-2 MSI (4.00)	K. Stelzer, A. Dille, D. Van der Zande	
2.0	2022/06/16		Inclusion of improved quality flagging (corrupt pixels, bottom reflection, cloud shadow)	K. Stelzer, A. Dille, J. Scholze D. Van der Zande, C. Lebreton	E.Böhm; V. Brando
3.0	2025/05/28	all	Update following format change of HROC products; EiS Nov 2025 Update to new Template	K. Stelzer, D. Van der Zande, C. Lebreton	E.Böhm; V. Brando

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GLOSSARY AND ABBREVIATIONS

ARC	Arctic region
BAL	Baltic Sea region
bb, BB	backscatter coefficient
bbp, BBP, BBp	particular backscatter coefficient
BC	Brockmann Consult
BLK	Black Sea region
C2RCC	Case 2 Regional CoastColour processor
CDSE	Copernicus Data Space Ecosystem
CF	Climate Forecast (convention for NetCDF)
CHL, CHL-a	Chlorophyll-a
CMEMS	Copernicus Marine Environment Monitoring Service
DGF	Direct Get File (FTP-like CMEMS service tool to download a NetCDF file)
ECMWF	European Centre for Medium Range Weather forecast
ESA	European Space Agency
HR	High Resolution
HR OC	High Resolution Ocean Colour Service
IBI	Iberian Sea region
L1C	Level 1C (here: Sentinel-2 L1C)
L2, L3, L4	Level 2, Level 3, Level 4 for data processing

L4D	L4 DINEOF water product
LR	Low Resolutions (> 300m)
MDS	Marine Data Store
MED	Mediterranean Sea region
MY	Multi Year
NetCDF	Network Common Data Form
NRT	Near Real Time
NWS	North-Western Shelf Region
OC	Ocean Colour
OGC	Open Geospatial Consortium
PU	Production Unit
PUM	Product User Manual
QI	Quality index
QUID	Quality Information Document
RBINS	Royal Belgian Institute for Nature
S2	Sentinel 2
SPM	Suspended particulate matter
TAC	Thematic Assembly Center
TUR	Turbidity
VITO	Flemish Institute for Technological Research NV

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Ocean Colour High Resolution products for the coastal zone of European Seas. It covers the products:

OCEANCOLOUR_ARC_BGC_HR_L3_NRT_009_201
OCEANCOLOUR_BAL_BGC_HR_L3_NRT_009_202
OCEANCOLOUR_NWS_BGC_HR_L3_NRT_009_203
OCEANCOLOUR_IBI_BGC_HR_L3_NRT_009_204
OCEANCOLOUR_MED_BGC_HR_L3_NRT_009_205
OCEANCOLOUR_BLK_BGC_HR_L3_NRT_009_206
OCEANCOLOUR_ARC_BGC_HR_L4_NRT_009_207
OCEANCOLOUR_BAL_BGC_HR_L4_NRT_009_208
OCEANCOLOUR_NWS_BGC_HR_L4_NRT_009_209
OCEANCOLOUR_IBI_BGC_HR_L4_NRT_009_210
OCEANCOLOUR_MED_BGC_HR_L4_NRT_009_211
OCEANCOLOUR_BLK_BGC_HR_L4_NRT_009_212

The products were introduced into Copernicus Marine OC-TAC with EiS May 2021 and cover products from 1-1-2020 onwards. The products cover all European Seas regions: ARC, BAL, NWS, IBI, MED, BLK within a 20km wide strip from the coast. With EiS November 2025, the format of the products changed from tile-wise products to full-region mosaics. The parameters delivered in different data sets are:

- Chlorophyll concentration (CHL)
- Turbidity (TUR)
- Suspended Particulate Matter concentration (SPM)
- Remote Sensing Reflectances (RRS)
- Particular Backscatter coefficients (BBP) of all bands
-

The products are provided on a daily basis and contain either daily averages (L3, P1D) or monthly averages (L4, P1M). A gap filled daily L4 product is provided every two months (L4, P1D). The products provide information for the coastal waters 20 km off the shoreline in 100 m spatial resolution. The coverage of the coastal waters in the 6 regions is shown in the maps in Figure 1. Full-region mosaics are available for L3 daily and gap filled L4 daily products and from EiS November 2025 onwards also for L4 monthly products.
The products are published on the Copernicus Marine dissemination server after automatic and human quality controls. Products are available on-line and disseminated through the Copernicus Marine Information System. Files downloaded are in NetCDF 4 format and follow CF-1.7 convention.

The processing chain and production system are described in the Quality Information Document (QUID)

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

The ocean colour coastal products cover 6 regional seas in Europe. The structure of the datasets within the Product Lines is the same for each region. Therefore, the following information is given for all Product Lines in common.

<table>
<tr><td>Product Lines</td><td colspan="5">OCEANCOLOUR_ARC_BGC_HR_L3_NRT_009_201
OCEANCOLOUR_BAL_BGC_HR_L3_NRT_009_202
OCEANCOLOUR_NWS_BGC_HR_L3_NRT_009_203
OCEANCOLOUR_IBI_BGC_HR_L3_NRT_009_204
OCEANCOLOUR_MED_BGC_HR_L3_NRT_009_205
OCEANCOLOUR_BLK_BGC_HR_L3_NRT_009_206

OCEANCOLOUR_ARC_BGC_HR_L4_NRT_009_207
OCEANCOLOUR_BAL_BGC_HR_L4_NRT_009_208
OCEANCOLOUR_NWS_BGC_HR_L4_NRT_009_209
OCEANCOLOUR_IBI_BGC_HR_L4_NRT_009_210
OCEANCOLOUR_MED_BGC_HR_L4_NRT_009_211
OCEANCOLOUR_BLK_BGC_HR_L4_NRT_009_212</td></tr>
<tr><td rowspan="6">Geographical coverage</td><td>ARC</td><td>58°N</td><td>90°N</td><td>55°W</td><td>32°E</td></tr>
<tr><td>BAL</td><td>53.25°N</td><td>66°N</td><td>9.25°E</td><td>31°E</td></tr>
<tr><td>NWS</td><td>48°N</td><td>62°N</td><td>20°W</td><td>13°E</td></tr>
<tr><td>IBI</td><td>26°N</td><td>48°N</td><td>20°W</td><td>5.61°W</td></tr>
<tr><td>MED</td><td>30°N</td><td>46°N</td><td>5.61°W</td><td>36.5°E</td></tr>
<tr><td>BLK</td><td>40°N</td><td>48°N</td><td>26.5°E</td><td>42°E</td></tr>
<tr><td>Variables</td><td colspan="5">Remote Sensing Reflectances (RRS)
Chlorophyll concentration (CHL)
Turbidity (TUR)
Suspended Particulate Matter concentration (SPM)
Particular Backscatter coefficients (BBP) of all bands</td></tr>
<tr><td>Product Type</td><td colspan="5">Near Real Time</td></tr>
<tr><td>Available time series</td><td colspan="5">L3 = 3 Days after acquisition
L4 = 15 Days after end of months (P1M) and 1month after respective period (P1D)</td></tr>
<tr><td>Temporal resolution</td><td colspan="5">Daily, Monthly</td></tr>
<tr><td>Update Frequency</td><td colspan="5">Daily, Monthly, every 2 months</td></tr>
<tr><td>Target delivery time</td><td colspan="5">daily at 11:59 pm UTC</td></tr>
<tr><td>Starting date of product availability</td><td colspan="5">01.01.2020</td></tr>
<tr><td>Horizontal resolution</td><td colspan="5">100m</td></tr>
<tr><td>Number of vertical levels</td><td colspan="5">1</td></tr>
</table>

Format	NetCDF4

b) Production System Description

The High Resolution Ocean Colour products (HROC) provide water quality parameters derived from Sentinel-2 Satellite data. The products cover the coastal zones (up to 20 km offshore) of all European Seas, namely NWS, BAL, IBI, MED, BLK, and the European part of the ARC region. The products are based on a multi-algorithm approach combining best suited algorithms for different water types. The resolution of the products is 100 m.

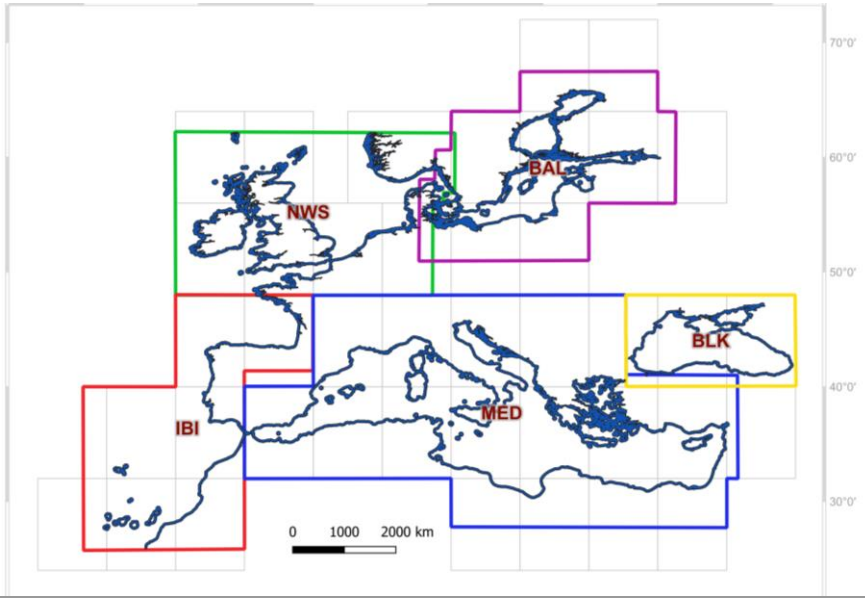
With EiS in November 2025 the products for the **BAL, NWS, IBI, MED**, and **BLK** regions are provided in regional mosaics to enhance accessibility of the products through the Copernicus Marine Tool and MyOcean Pro viewer and ensure the proper functioning of all user interfaces, such as the subset tool. The region wide mosaic layout is shown as colorized polygons in the first figure in the table below. The projection used is equi-rectangular geographic. The products that were provided as tiles will be frozen. Thus they are still accessible, but the NRT service stops in November 2025. Those products were provided in 6x8° tiles, shown in light grey in the same figure.

For **ARC**, the products are provided as tiles to manage file sizes. The tiling uses 6° × 8° blocks similar to UTM zones. ARC products are provided in Lambert Azimuthal Equal Area projection (LAEA, EPSG:6931 or EASE2). ARC products will be continued to be delivered in tiles.

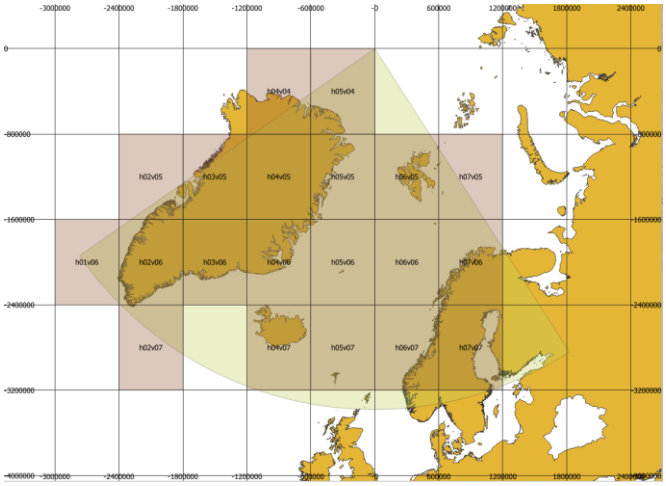
The HR-OC processing system is deployed on the Copernicus Data Space Ecosystem (CDSE) with the production control element being hosted within the infrastructure of Brockmann Consult.

Further details on the processing chain and production system are provided in the QUID. Short facts are provided in the overview table below:

BAL: 9.25°E --> 31°E; 53.25°N --> 66°N
NWS: 20°W --> 13°E; 48°N --> 62°N
IBI: 20°W --> 5.61°W; 26°N --> 48°N
MED: 5.61°W --> 36.5°E; 30°N --> 46°N
BLK: 26.5°E --> 42°E; 40°N --> 48°N



ARC: 55°W --> 32°E; 58°N --> 90°N



System Name	HROC production system
Processor version	V01
Upstream products	Sentinel-2 MSI A/B (production baseline v. 02.04 until 24.01.2022) Sentinel-2 MSI A/B (production baseline v.04.00 since 25.01.2022) Sentinel-2 MSI A/B (production baseline v.5.11 since 23.07.2023) Sentinel-2 MSI B/C (replacement of Sentinel -2 A with Sentinel-2C on 23.01.2025)
Horizontal Resolution	**IBI, MED,** and **BLK**: 1/1080 degree N-S and 1/756 degree W-E (ratio 0.7) (=close to 100 m in each direction) **NWS** and **BAL**: 1/1080 degree N-S and 1/572 degree W-E (ratio 0.53) (=close to 100 m in each direction) **ARC**: 100 m
Coordinate systems	**BAL, NWS, IBI, MED, BLK**: geographic lat/lon grid (EPSG:4326) **ARC**: Lambert Azimuthal Equal Area projection (LAEA, EPSG:6931 or EASE2, region ARC)
Frequency	Daily delivery of all L3 products, but not daily coverage depending on satellite overpasses
Pixel identification	Idepix-MSI (SNAP) and spectral tests
Atmospheric correction algorithms	HR-OC L2R processor (C2RCC and ACOLITE combined in a merging approach)
In-water retrieval	HR-OC L2W processor
Product levels	Level 3 daily (I3-P1D) - NRT products, delivered daily, 1-3 days after acquisition Level 4 daily (I4-P1D) – daily gap-filled data, delivered every 2 months Level 4 monthly (I4-P1M) – monthly averages, delivered every month

Product dissemination format	NetCDF data sets for optics, transparency and geophysics in a tile system covering 6x8° sub-regions until November 2025. Starting in November 2025, the tile mosaics are provided as regional-wide mosaics, delivered in NetCDF datasets per day: one for **optics** and one for **turbidity, suspended matter and chlorophyll concentration.** This relates to the regions **BAL, NWS, IBI, MED, BLK.** **ARC** continues to be delivered on a tile basis.

c) Dataset details

The HR-OC datasets are provided as **Level 3** daily, NRT products and as **Level 4** datasets (gap-filled daily and monthly averages). There is a table characterizing the Level 3 datasets and one table describing the Level 4 datasets. The content of the Product Lines for the regions **BAL**, **NWS**, **IBI**, **MED** and **BLK** is the same. Therefore, they are described commonly in one table. The **ARC** differs in some aspects and is therefore described in a separate table.

Product Lines	OCEANCOLOUR_**BAL**_BGC_HR_**L3**_NRT_009_202 OCEANCOLOUR_**NWS**_BGC_HR_**L3**_NRT_009_203 OCEANCOLOUR_**IBI**_BGC_HR_**L3**_NRT_009_204 OCEANCOLOUR_**MED**_BGC_HR_**L3**_NRT_009_205 OCEANCOLOUR_**BLK**_BGC_HR_**L3**_NRT_009_206
Geographical coverage	European regional Seas (BAL, NWS, IBI, MED, BLK)
Datasets: cmems_obs_oc_<region>_bgc-tur-spm-chl_nrt_l3-hr-mosaic_**P1D**-m *(available since Jan 2020)* cmems_obs_oc_<region>_bgc_optics_nrt_l3-hr-mosaic_**P1D**-m *(start in Nov 2025)* cmems_obs_oc_<region>_bgc_optics_nrt_l3-hr_**P1D**-m *(until Nov 2025)* cmems_obs_oc_<region>_bgc_transp_nrt_l3-hr_**P1D**-m *(until Nov 2025)* cmems_obs_oc_<region>_bgc_geophy_nrt_l3-hr_**P1D**-m *(until Nov 2025)*	
Product Type	Observation
Update frequency	Daily
Available time series	up to 3 days after acquisition (most of the products are delivered within 1 day after acquisition
Target delivery time	23:59 UTC
Temporal resolution	Daily mean
Starting date of product availability	01.01.2020
Geographic coordinate system	Geographic lat/lon grid EPSG: 4326

Dataset : cmems_obs_oc_<region>_bgc-tur-spm-chl_nrt_l3-hr-mosaic_P1D contains regional mosaics covering full region for variables turbidity, suspended matter concentration and chlorophyll- concentration.
TUR [FNU] Turbidity derived from MSI L2R using HR-OC L2W processor sea_water_turbidity
SPM [g m-3]

Suspended Particulate Matter derived from MSI L2R using HR-OC L2W processor mass_concentration_of_suspended_matter_in_sea_water
CHL [mg m-3] Chlorophyll-a concentration derived from MSI L2R using HR-OC L2W processor mass_concentration_of_chlorophyll_a_in_sea_water
Dataset : cmems_obs_oc_<region>_bgc_optics_nrt_l3-hr-mosaic_P1D contains the leaving remote sensing reflectances (RRS) in 8 spectral bands plus the uncertainty measure per for each wavelength. It further contains the backscatter (BBP) for the same spectral bands. Wavelengths in nm of spectral bands (xxx in names below): 443, 492, 560, 665, 704, 740, 783, 865. Provision started with EiS November 2025 and is replacing tile based optics products
RRSxxx [sr-1] remote sensing Reflectances for 8 spectral wavelengths surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air xxx = Wavelengths in 443, 492, 560, 665, 704, 740, 783, 865 nm
RRS_UNCxxx [sr-1] Relative uncertainty of RRS at dedicated wavelength from MSI L2R using HR-OC L2R processor xxx = Wavelengths in 443, 492, 560, 665, 704, 740, 783, 865 nm
BBPxxx [m-1] Backscattering at xxx nm from MSI L2R using HR-OC L2R processor volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water xxx = Wavelengths in 443, 492, 560, 665, 704, 740, 783, 865 nm
Dataset: cmems_obs_oc_<region>_bgc_optics_nrt_l3-hr_P1D-m Contains the water leaving remote sensing reflectances (RRS) in 8 spectral bands plus the uncertainty measure per for each wavelength. It further contains the backscatter (BBP) for the same spectral bands. Wavelengths in nm of spectral bands. <region> = BAL, NWS, IBI, MED, BLK Provided until November 2025
RRSxxx [sr-1] remote sensing Reflectances for 8 spectral wavelengths surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air xxx = Wavelengths in 443, 492, 560, 665, 704, 740, 783, 865 nm
RRS_UNCxxx [sr-1] Relative uncertainty of RRS at dedicated wavelength from MSI L2R using HR-OC L2R processorxxx = Wavelengths in 443, 492, 560, 665, 704, 740, 783, 865 nm
BBPxxx [m-1] Backscattering at xxx nm from MSI L2R using HR-OC L2R processor volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water xxx = Wavelengths in 443, 492, 560, 665, 704, 740, 783, 865 nm
Dataset: cmems_obs_oc_<region>_bgc_transp_nrt_l3-hr_P1D-m Contains Suspended Matter concentration and Turbidity of the water column, including Quality Index per variable. Provided until November 2025
SPM [g m-3] Suspended Particulate Matter derived from MSI L2R using HR-OC L2W processor mass_concentration_of_suspended_matter_in_sea_water
TUR [FNU] Turbidity derived from MSI L2R using HR-OC L2W processor

sea_water_turbidity
SPM_QI [%] Quality Index for suspended particulate matter derived from comparison with climatology SPM_QI relative_percentage_variation_index
TUR_QI [%] Quality Index for turbidity derived from comparison with climatology TUR_QI relative_percentage_variation_index
Dataset : cmems_obs_oc_<region>_bgc_geophy_nrt_l3-hr_P1D-m Contains the Chlorophyll-a concentration in water. Provided until November 2025
CHL [mg m-3] Chlorophyll-a concentration derived from MSI L2R using HR-OC L2W processor mass_concentration_of_chlorophyll_a_in_sea_water

Product Lines	OCEANCOLOUR_**ARC**_BGC_HR_**L3**_NRT_009_207
Geographical coverage	European Waters of ARC region. The region is provided in 6°x8° tiles.
Datasets: cmems_obs_oc_ARC_bgc_optics_nrt_l3-hr_P1D-m cmems_obs_oc_ARC_bgc_transp_nrt_l3-hr_P1D-m cmems_obs_oc_ARC_bgc_geophy_nrt_l3-hr_P1D-m	
Product Type	Observation
Update frequency	Daily
Available time series	up to 3 days after acquisition (most of the products are delivered 1 day after acquisition
Target delivery time	23:59 UTC
Temporal resolution	Daily mean
Starting date of product availability	01.01.2020
Geographic coordinate system	Lambert Azimuthal Equal Area projection (LAEA, EPSG:6931 or EASE2), provided in 6x8° tiles

Dataset: cmems_obs_oc_ARC_bgc_optics_nrt_l3-hr_P1D-m Contains the water leaving remote sensing reflectances (RRS) in 8 spectral bands plus the uncertainty measure per for each wavelength. It further contains the backscatter (BBP) for the same spectral bands.
RRSxxx [sr-1] remote sensing Reflectances for 8 spectral wavelengths surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air xxx = Wavelengths in 443, 492, 560, 665, 704, 740, 783, 865 nm
RRS_UNCxxx [sr-1] Relative uncertainty of RRS at dedicated wavelength from MSI L2R using HR-OC L2R processor xxx = Wavelengths in 443, 492, 560, 665, 704, 740, 783, 865 nm
BBPxxx [m-1] Backscattering at xxx nm from MSI L2R using HR-OC L2R processor volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water xxx = Wavelengths in 443, 492, 560, 665, 704, 740, 783, 865 nm
Dataset: cmems_obs_oc_ARC_bgc_transp_nrt_l3-hr_P1D-m Contains Suspended Matter concentration and Turbidity of the water column, including Quality Index per variable.
SPM [g m-3] Suspended Particulate Matter derived from MSI L2R using HR-OC L2W processor mass_concentration_of_suspended_matter_in_sea_water
TUR [FNU] Turbidity derived from MSI L2R using HR-OC L2W processor sea_water_turbidity
SPM_QI [%] Quality Index for suspended particulate matter derived from comparison with climatology SPM_QI relative_percentage_variation_index
TUR_QI [%] Quality Index for turbidity derived from comparison with climatology

TUR_QI relative_percentage_variation_index
Dataset : cmems_obs_oc_ARC_bgc_geophy_nrt_l3-hr_P1D-m Contains the Chlorophyll-a concentration in water. Provided until November 2025
CHL [mg m-3] Chlorophyll-a concentration derived from MSI L2R using HR-OC L2W processor mass_concentration_of_chlorophyll_a_in_sea_water

Product Lines	OCEANCOLOUR_**BAL**_BGC_HR_**L4**_NRT_009_208 OCEANCOLOUR_**NWS**_BGC_HR_**L4**_NRT_009_209 OCEANCOLOUR_**IBI**_BGC_HR_**L4**_NRT_009_210 OCEANCOLOUR_**MED**_BGC_HR_**L4**_NRT_009_211 OCEANCOLOUR_**BLK**_BGC_HR_**L4**_NRT_009_212
Geographical coverage	European regional Seas (BAL, NWS, IBI, MED, BLK)
Datasets: cmems_obs_oc_<region>_bgc_tur_spm_chl_nrt_l4-hr-mosaic_**P1D**-m cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr-mosaic_**P1D**-m cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr_**P1D**-m *(until Nov 2025)* cmems_obs_oc_<region>_bgc_transp_nrt_l4-hr_**P1D**-m *(until Nov 2025)* cmems_obs_oc_<region>_bgc_geophy_nrt_l4-hr_**P1D**-m *(until Nov 2025)* cmems_obs_oc_<region>_bgc_tur_spm_chl_nrt_l4-hr-mosaic_**P1M**-m cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr -mosaic_**P1M**-m cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr_**P1M**-m *(until Nov 2025)* cmems_obs_oc_<region>_bgc_transp_nrt_l4-hr_**P1M**-m *(until Nov 2025)* cmems_obs_oc_<region>_bgc_geophy_nrt_l4-hr_**P1M**-m *(until Nov 2025)*	
Product Type	Observation
Update frequency	Daily gap-filled (P1D): every 2 months Monthly averages (P1M): monthly
Available time series	Daily gap-filled (P1D): 1 Month after end of period, every 2 months Monthly averages (P1M): 15 days after end of the month
Target delivery time	23:59 UTC
Temporal resolution	Monthly (P1M) and Daily (P1D)
Starting date of product availability	01.01.2020
Geographic coordinate system	Geographic lat/lon grid EPSG: 4326

Dataset : cmems_obs_oc_<region>_bgc_tur_spm_chl_nrt_l4-hr-mosaic_P1D-m contains regional mosaics covering full region for gap-filled variables turbidity, suspended particulate matter concentration and chlorophyll-a concentration. <region> = BAL, NWS, IBI, MED, BLK
TUR [FNU] Turbidity derived from MSI L2R using HR-OC L2W processor sea_water_turbidity
SPM [g m-3] Suspended Particulate Matter derived from MSI L2R using HR-OC L2W processor mass_concentration_of_suspended_matter_in_sea_water
CHL [mg m-3] Chlorophyll-a concentration derived from MSI L2R using HR-OC L2W processor mass_concentration_of_chlorophyll_a_in_sea_water
Dataset : cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr-mosaic_P1D-m contains the backscatter (BBP) at 443nm, gap-filled. Products are provided as regional-wide mosaics. Provision started with EiS November 2025 <region> = BAL, NWS, IBI, MED, BLK

Provision started with EiS November 2025 and is replacing tile based optics products
BBP443 [m-1] Backscattering at 443 nm from MSI L2R using HR-OC L2R processor volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water
Dataset: cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr_P1D-m Contains contains the gap-filled backscatter (BBP) at 443nm. <region> = BAL, NWS, IBI, MED, BLK Provided until November 2025
BBP443 [m-1] Backscattering at 443nm from MSI L2R using HR-OC L2R processor volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water
Dataset: cmems_obs_oc_<region>_bgc_transp_nrt_l34hr_P1D-m Contains gap-filled Suspended Matter concentration and Turbidity of the water column, including Quality Index per variable. Provided until November 2025 <region> = BAL, NWS, IBI, MED, BLK Provided until November 2025
SPM [g m-3] Suspended Particulate Matter derived from MSI L2R using HR-OC L2W processor mass_concentration_of_suspended_matter_in_sea_water
TUR [FNU] Turbidity derived from MSI L2R using HR-OC L2W processor sea_water_turbidity
Dataset : cmems_obs_oc_<region>_bgc_geophy_nrt_l4-hr_P1D-m Contains the gap-filled Chlorophyll-a concentration in water. Provided until November 2025 <region> = BAL, NWS, IBI, MED, BLK Provided until November 2025
CHL [mg m-3] Chlorophyll-a concentration derived from MSI L2R using HR-OC L2W processor mass_concentration_of_chlorophyll_a_in_sea_water
Dataset : cmems_obs_oc_<region>_bgc_tur_spm_chl_nrt_l4-hr-mosaic_P1M-m Contains regional mosaics covering full region for variables turbidity, suspended matter concentration and chlorophyll- concentration. Monthly averages.
TUR [FNU] Turbidity derived from MSI L2R using HR-OC L2W processor sea_water_turbidity
SPM [g m-3] Suspended Particulate Matter derived from MSI L2R using HR-OC L2W processor mass_concentration_of_suspended_matter_in_sea_water
CHL [mg m-3] Chlorophyll-a concentration derived from MSI L2R using HR-OC L2W processor mass_concentration_of_chlorophyll_a_in_sea_water
Dataset : cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr-mosaic_P1M-m Contains the monthly averages of backscatter (BBP) at 443nm. Products are provided as regional-wide mosaics. Provision started with EiS November 2025 and is replacing the tile-based optics product
BBP443 [m-1] Backscattering at 443 nm from MSI L2R using HR-OC L2R processor volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water

Dataset: cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr_P1M-m contains the monthly averages of backscatter (BBP) at 443nm. <region> = BAL, NWS, IBI, MED, BLK Provided until November 2025
BBP443 [m-1] Backscattering at 443 nm from MSI L2R using HR-OC L2R processor (monthly averages) volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water
Dataset: cmems_obs_oc_<region>_bgc_transp_nrt_l34hr_P1M-m Contains monthly averages of Suspended Matter concentration and Turbidity of the water column, including Quality Index per variable. Provided until November 2025
SPM [g m-3] Suspended Particulate Matter derived from MSI L2R using HR-OC L2W processor (monthly averages) mass_concentration_of_suspended_matter_in_sea_water
TUR [FNU] Turbidity derived from MSI L2R using HR-OC L2W processor sea_water_turbidity
SPM_count [1] Number of daily observations for Suspended Particulate Matter
TUR_count [1] Number of daily observations for Turbidity
SPM_error [g m-3] Standard deviation of observations of Suspended Particulate Matter mass_concentration_of_suspended_matter_in_sea_water standard_error
TUR_error [FNU] Standard deviation of observations of Turbidity Sea_water_turbidity standard_error
Dataset : cmems_obs_oc_<region>_bgc_geophy_nrt_l4-hr_P1M-m Contains the monthly averages of Chlorophyll-a concentration in water. Provided until November 2025
CHL [mg m-3] Chlorophyll-a concentration derived from MSI L2R using HR-OC L2W processor mass_concentration_of_chlorophyll_a_in_sea_water
CHL_count [1] Number of daily observations for Chlorophyll-a concentration
CHL_error [mg m-3] Standard deviation of observations of Chlorophyll-a concentration mass_concentration_of_chlorophyll_a_in_sea_water standard_error

Product Lines	CEANCOLOUR_**ARC**_BGC_HR_**L4**_NRT_009_207
Geographical coverage	European Waters of ARC region. The region is provided in 6°x8° tiles.
Datasets: cmems_obs_oc_ARC_bgc_optics_nrt_l4-hr_**P1D**-m cmems_obs_oc_ARC_bgc_transp_nrt_l4-hr_**P1D**-m cmems_obs_oc_ARC_bgc_geophy_nrt_l4-hr_**P1D**-m cmems_obs_oc_ARC_bgc_optics_nrt_l4-hr_**P1M**-m	

cmems_obs_oc_ARC_bgc_transp_nrt_l4-hr_**P1M**-m cmems_obs_oc_ARC_bgc_geophy_nrt_l4-hr_**P1M**-m	
Product Type	Observation
Update frequency	Daily gap-filled (P1D): Bi-monthly Monthly averages (P1M): monthly
Available time series	Daily gap-filled (P1D): 1 Month after end of period, every 2 months Monthly averages (P1M): 15 days after end of the month
Target delivery time	23:59 UTC
Temporal resolution	Monthly (P1M) and Daily (P1D)
Starting date of product availability	01.01.2020
Geographic coordinate system	Lambert Azimuthal Equal Area projection (LAEA, EPSG:6931 or EASE2), provided in 6x8° tiles

Dataset: cmems_obs_oc_ARC_bgc_optics_nrt_l4-hr_P1D-m Contains the gap-filled backscatter (BBP) at 443nm.
BBP443 [m-1] Backscattering at 443nm from MSI L2R using HR-OC L2R processor volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water
Dataset: cmems_obs_oc_ARC_bgc_transp_nrt_l34hr_P1D-m Contains gap-filled Suspended Matter concentration and Turbidity of the water column, including Quality Index per variable.
SPM [g m-3] Suspended Particulate Matter derived from MSI L2R using HR-OC L2W processor mass_concentration_of_suspended_matter_in_sea_water
TUR [FNU] Turbidity derived from MSI L2R using HR-OC L2W processor sea_water_turbidity
Dataset : cmems_obs_oc_ARC_bgc_geophy_nrt_l4-hr_P1D-m Contains the gap-filled Chlorophyll-a concentration in water.
CHL [mg m-3] Chlorophyll-a concentration derived from MSI L2R using HR-OC L2W processor mass_concentration_of_chlorophyll_a_in_sea_water

d) Additional information on some parameters

The Level 3 and Level 4 raster products of the regions BAL, NWS, IBI, MED, and BLK are in geographic lat-lon grid (EPSG:4326). The products of region ARC are in polar Lambertian Azimuthal Equal Area (LAEA) projection (EPSG:6931, EASE2).
The target resolution of the grid is 100m. The grid of the geographic projection has grid cells rectangular in degree to reveal pixels almost square in meters depending on a reference latitude of each region. The grid (pixel corner coordinates) is aligned to integer degree meridians and latitudes. The grid divides 1-degree cells into 1080 * 756 pixels (ratio 0.7) for IBI, MED, and BLK, and into 1080 * 572 pixels (ratio 0.53) for NWS and BAL. This means that the pixels are square with 100 m stepping close to 45 North reference latitude for IBI, MED, and BLK, and they are metric squares close to 58 North reference latitude for NWS and BAL.

With EiS November 2025, the delivery format of the regions BAL, NWS, IBI, MED and BLK has changed. Starting in November 2025, the products are provided as regional mosaics, while the ARC region is continued to being provided in tiles. The tiles-based products for BAL, NWS, IBI, MED and BLK are still available in the catalogue, but stop in November 2025.

e) Processing information

The incoming satellite data (Sentienl-2 MSI) are processed on a daily basis. If input products arrive in time (appr. 5 hours after acquisition), the processing of L3 products lasts until lunch time of the following day. Products are disseminated at 15UTC that day if all input products have been processed to fill the provided mosaics. If input products are on delay or production process takes longer, the dissemination is postponed automatically to 21UTC. If input products are still missing, the production is paused up to 3 days after the day of acquisition.
A visual inspection of all mosaics is performed, and the production process is stopped in case of issues. Issues are solved within 3days after acquisition. If this is not possible, users are informed about late delivery.
Monthly average products are generated from the daily mosaics and provided 15 days after the last day of the month. However, if all input products are available earlier, the products are provided before.
Daily gap-filled products are generated every 2 months using a time period of 3 months as input (½months before and ½ month after the 2 months-period). The gap-filled daily products are delivered 1 months after the last day of the covered period.

Product disseminating schedule :

Type of product	temporal resolution	dissemination day	frequency
observation L3D NRT	daily	Up to 3 days after acquisition	daily
observation L4M	monthly	15 days after end of month	monthly
observation L4D	daily	1 month after end of period	every 2 months

f) Catalogue updating

Products are uploaded to the catalogue. The catalogue holds all products that have been uploaded since EiS of the respective product. The former tile-based products are kept in the catalogue but not continued. Each new product file is a new entry in the catalogue.

g) Time averaging

Dataset frequency	Time averaging	delivery time
monthly	monthly means over the calendar month (first to last day of the month)	15D after last day of month (often 3D after acquisition)
daily	daily means over a day: overlapping satellite orbits are averaged. Areas are mainly covered only by one overpass per day.	3D after day of acquisition (often 1D after acquisition)

h) Product use and impact

The ocean colour coastal products are used for detailed investigation of water quality parameters close to the coast. Due to its high resolution of 100m, fine-scale structures can be detected for the different parameters. The water quality parameters chlorophyll concentration, turbidity or suspended sediment concentration can be directly used for spatial and temporal analyses of the coastal region of interest.
In addition, own algorithms can be applied and tested as the products also provide the water leaving reflectances as base information. They are stored in the OPTICS products.
With the provision of region-wide mosaics (starting with EiS Nov 2025), the products are available via all Copernicus Marine tools. They can be downloaded as full regional product or as subset of a certain bay or region of interest.
A first step to investigate the products is to visualize the data via the MyOcean viewer and select desired products for download. The MyOcean viewer Pro is accessible via this link: https://marine.copernicus.eu/access-data/myocean-viewer
Look for the respective data set in "Add Layer" and you can browse through the data. With the download icon you are directed back to the CMEMS catalogue, directly to the respective product.

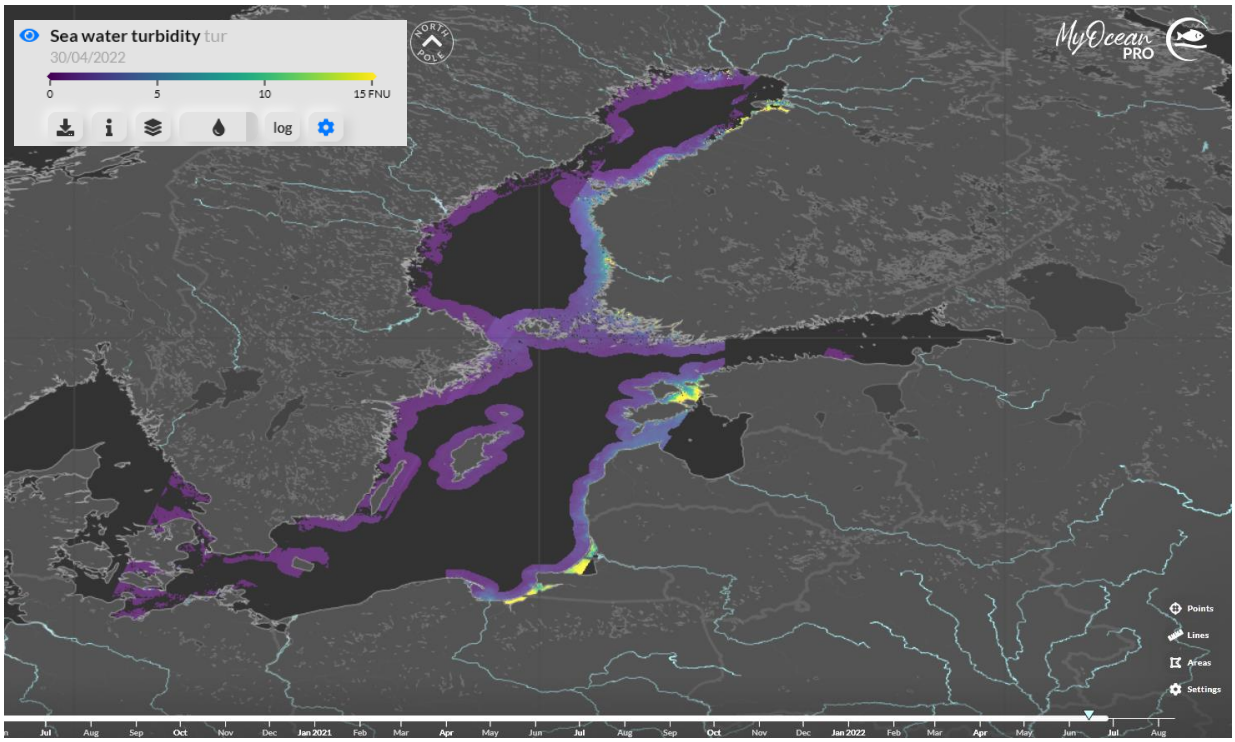


Figure 1: MyOcean Viewer with TUR displayed for the mosaicked BALTIC region

For the HR products, five parameters are available as full regional products (CHL, TUR, TSM, RRS, BBP) for L3 daily and CHL, TUR, TSM, BBP443 for L4D gap-filled and L4M monthly products. They can be visualized in the MyOcean Viewer.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Dataset names	Product names
cmems_obs_oc_<region>_bgc_optics_nrt_l3-hr-mosaic_P1D-m	**${date}_P1D_CMEMS_HROC_L3-optics_<region>__100m-v01.nc**
cmems_obs_oc_<region>_bgc_tur-spm-chl_nrt_l3-hr-mosaic_P1D-m	**${date}_P1D_CMEMS_HROC_L3-tur_spm_chl_<region>__100m-v01.nc**
cmems_obs_oc_ARC_bgc_optics_nrt_l3-hr_P1D-m	**${date}_P1D_CMEMS_HROC_L3-optics_ARC_<Tile>_100m-v01.nc**
cmems_obs_oc_ARC_bgc_transp_nrt_l3-hr_P1D-m	**${date}_P1D_CMEMS_HROC_L3-transp_ARC_<Tile>_100m-v01.nc**
cmems_obs_oc_ARC_bgc_geophy_nrt_l3-hr_P1D-m	**${date}_P1D_CMEMS_HROC_L3-geophy_ARC_<Tile>_100m-v01.nc**
cmems_obs_oc_<region>_bgc_optics_nrt_l3-hr_P1D-m	${date}_P1D_CMEMS_HROC_L3-optics_<region>_<tile>_100m-v01.nc
cmems_obs_oc_<region>_bgc_transp_nrt_l3-hr_P1D-m	${date}_P1D_CMEMS_HROC_L3-transp_<region>_<tile>_100m-v01.nc
cmems_obs_oc_<region>_bgc_geophy_nrt_l3-hr_P1D-m	${date}_P1D_CMEMS_HROC_L3-geophy_<region>_<tile>_100m-v01.nc
cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr-mosaic_P1M-m	**${date}_P1M_CMEMS_HROC_L4-optics_<region>__100m-v01.nc**

cmems_obs_oc_<region>_bgc_tur-spm-chl_nrt_l4-hr-mosaic_P1M-m	**${date}_P1M_CMEMS_HROC_L4-tur-spm-chl_<region>__100m-v01.nc**
cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr-mosaic_P1D-m	**${date}_P1D_CMEMS_HROC_L4-optics_<region>__100m-v01.nc**
cmems_obs_oc_<region>_bgc_tur-spm-chl_nrt_l4-hr-mosaic_P1D-m	**${date}_P1D_CMEMS_HROC_L4-tur-spm-chl_<region>__100m-v01.nc**
cmems_obs_oc_ARC_bgc_optics_nrt_l4-hr_P1D-m	**${date}_P1D_CMEMS_HROC_L4-optics_ARC_<Tile>_100m-v01.nc**
cmems_obs_oc_ARC_bgc_transp_nrt_l4-hr_P1D-m	**${date}_P1D_CMEMS_HROC_L4-transp_ARC_<Tile>_100m-v01.nc**
cmems_obs_oc_ARC_bgc_geophy_nrt_l4-hr_P1D-m	**${date}_P1D_CMEMS_HROC_L4-geophy_ARC_<Tile>_100m-v01.nc**
cmems_obs_oc_ARC_bgc_optics_nrt_l4-hr_P1M-m	**${date}_P1M_CMEMS_HROC_L4-optics_ARC_<Tile>_100m-v01.nc**
cmems_obs_oc_ARC_bgc_transp_nrt_l4-hr_P1M-m	**${date}_P1M_CMEMS_HROC_L4-transp_ARC_<Tile>_100m-v01.nc**
cmems_obs_oc_ARC_bgc_geophy_nrt_l4-hr_P1M-m	**${date}_P1M_CMEMS_HROC_L4-geophy_ARC_<Tile>_100m-v01.nc**
cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr_P1D-m	${date}_P1D_CMEMS_HROC_L4-optics_<region>_<Tile>_100m-v01.nc
cmems_obs_oc_<region>_bgc_transp_nrt_l4-hr_P1D-m	${date}_P1D_CMEMS_HROC_L4-transp_<region>_<Tile>_100m-v01.nc
cmems_obs_oc_<region>_bgc_geophy_nrt_l4-hr_P1D-m	${date}_P1D_CMEMS_HROC_L4-geophy_<region>_<Tile>_100m-v01.nc
cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr_P1M-m	${date}_P1M_CMEMS_HROC_L4-optics_<region>_<Tile>_100m-v01.nc
cmems_obs_oc_<region>_bgc_transp_nrt_l4-hr_P1M-m	${date}_P1M_CMEMS_HROC_L4-transp_<region>_<Tile>_100m-v01.nc
cmems_obs_oc_<region>_bgc_geophy_nrt_l4-hr_P1M-m	${date}_P1M_CMEMS_HROC_L4-geophy_<region>_<Tile>_100m-v01.nc

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1 FILE SIZE

DATASET NAME[1]	FILE NAME	DIMENSION [GB]	
		compressed	decompressed
cmems_obs_oc_ARC_bgc_optics_nrt_l3-hr_P1D-m	${date}_P1D_CMEMS_HROC_L3-optics_ARC_<Tile>_100m-v1.0.nc	7.6	12.4
cmems_obs_oc_ARC_bgc_transp_nrt_l3-hr_P1D-m	${date}_P1D_CMEMS_HROC_L3-transp_ARC_<Tile>_100m-v1.0.nc	7.6	8.4
cmems_obs_oc_ARC_bgc_geophy_nrt_l3-hr_P1D-m	${date}_P1D_CMEMS_HROC_L3-geophy_ARC_<Tile>_100m-v1.0.nc	7.6	8
cmems_obs_oc_<region>_bgc_tur-spm-chl_nrt_l3-hr-mosaic_P1D-m	${date}_P1D_CMEMS_HROC_L3-tur_spm_chl_<region>_100m-v1.1.nc	0.05 - 0.1	1 - 6.5
cmems_obs_oc_<region>_bgc_optics_nrt_l3-hr-mosaic_P1D-m	${date}_P1D_CMEMS_HROC_L4-optics_<region>_<Tile>_100m-v1.0.nc	0.2-0.8	9-15
cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr-mosaics_P1P-m	${date}_P1P_CMEMS_HROC_L4-geophy_<region>_<Tile>_100m-v1.0.nc	0.02 – 0.03	0.3-2
cmems_obs_oc_<region>_bgc_optics_nrt_l4-hr-mosaics_P1M-m	${date}_P1M_CMEMS_HROC_L4-geophy_<region>_<Tile>_100m-v1.0.nc	0.03 – 0.2	0.3-2
cmems_obs_oc_<region>_bgc_tur-spm-chl_nrt_l4-hr-mosaic_P1D	${date}_P1D_CMEMS_HROC_L4-tur_spm_chl_<region>_100m-v1.1.nc	0.05 - 0.1	1 - 6.5
cmems_obs_oc_<region>_bgc_tur-spm-chl_nrt_l4-hr-mosaic_P1M	${date}_P1M_CMEMS_HROC_L4-tur_spm_chl_<region>_100m-v1.1.nc	0.1 – 0.5	1.1-6

[1] For full-region mosaics, the range of file sizes is given. They differ between regions and dates, therefore ranges are provided. <region> = BAL, BLK, MED, IBI, NWS. ARC is not provided in full region.

4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for representative datasets.

5) REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-HR-OC-QUID-009-201to2012-v4.0.pdf

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Doerffer, R., Schiller, H., Jürgen Fischer, Rene Preusker, and Marc Bouvet, (2008): The Impact of Sun Glint on the Retrieval of Water Parameters and Possibilities for the Correction of MERIS Scenes. In Proceedings of the 2nd MERIS/(A) ATSR User Workshop, ESA SP-666,(ESA, Frascati, taly,2008) https://earth.esa.int/pub/ESA_DOC/meris_workshop_2008/papers%20/o34_doer.pdf.

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Lavigne H. & Van der Zande D. & Ruddick K. & Cardoso Dos Santos J. & Gohin F. & Brotas V. & Kratzer S. Quality-control tests for OC4, OC5 and NIR-red satellite chlorophyll-a algorithms applied to coastal waters (2021) Remote Sensing of Environment, Vol. 255 p. 112237.

Lee Z., Kendall L., Arnone R. (2002) Deriving inherent optical properties from water color: a multiband quasianalytical algorithm for optically deep waters. Applied Optics, 41(27), 5755-5772.

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O'Reilly, J.E., Werdell, P.J., (2019): Chlorophyll algorithms for ocean color sensors-OC4, OC5 & OC6. Remote Sensing of Environment. 229, 32–47.

Vanhellemont, Q. (2019): Adaptation of the Dark Spectrum Fitting Atmospheric Correction for Aquatic Applications of the Landsat and Sentinel-2 Archives. Remote Sensing of Environment 225: 175–192.

Vanhellemont, Q., and Ruddick, K.G (2014b): Turbid Wakes Associated with Offshore Wind Turbines Observed with Landsat 8. Remote Sens. Environ. 145: 105–15.

Vanhellemont, Q., and K. Ruddick. 2015. Advantages of High Quality SWIR Bands for Ocean Colour Processing: Examples from Landsat-8. Remote Sensing of Environment 161 (May): 89–106. https://doi.org/10.1016/j.rse.2015.02.007.

6) ANNEX

For dataset cmems_obs_oc_<region>_bgc_tur-spm-chl_nrt_l3-hr-mosaic_P1D-m
(example of BAL region)

```
netcdf \20250511_P1D_CMEMS_HROC_L3-tur-spm-chl_BAL__100m-v01.9 {
dimensions:
	time = 1 ;
	lat = 14040 ;
	lon = 12594 ;
variables:
	float SPM(time, lat, lon) ;
		string SPM:long_name = "Suspended Particulate Matter derived from MSI L2R using HR-OC
L2W processor" ;
		string SPM:standard_name = "mass_concentration_of_suspended_matter_in_sea_water" ;
		string SPM:units = "g m-3" ;
		SPM:_FillValue = NaNf ;
		SPM:valid_min = 0.f ;
		string SPM:coordinates = "lat lon" ;
	float TUR(time, lat, lon) ;
		string TUR:long_name = "Turbidity derived from MSI L2R using HR-OC L2W processor" ;
		string TUR:standard_name = "sea_water_turbidity" ;
		string TUR:units = "FNU" ;
		TUR:_FillValue = NaNf ;
		TUR:valid_min = 0.f ;
		string TUR:coordinates = "lat lon" ;
	float CHL(time, lat, lon) ;
		string CHL:long_name = "Chlorophyll-a concentration derived from MSI L2R using HR-OC
L2W processor" ;
		string CHL:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
		string CHL:units = "mg m-3" ;
		CHL:_FillValue = NaNf ;
		CHL:valid_min = 0.f ;
		string CHL:coordinates = "lat lon" ;
	int crs ;
		string crs:wkt = "GEOGCS[\"WGS 84\", \n  DATUM[\"World Geodetic System 1984\", \n
SPHEROID[\"WGS 84\", 6378137.0, 298.257223563, AUTHORITY[\"EPSG\",\"7030\"]], \n
AUTHORITY[\"EPSG\",\"6326\"]], \n  PRIMEM[\"Greenwich\", 0.0, AUTHORITY[\"EPSG\",\"8901\"]], \n
UNIT[\"degree\", 0.017453292519943295], \n  AXIS[\"Geodetic longitude\", EAST], \n  AXIS[\"Geodetic
latitude\", NORTH], \n  AUTHORITY[\"EPSG\",\"4326\"]]" ;
		string crs:i2m = "0.0017470300489168414,0.0,0.0,-9.25925925925926E-
4,8.998951781970646,66.0" ;
	double time(time) ;
		string time:standard_name = "time" ;
		string time:axis = "T" ;
		string time:calendar = "Gregorian" ;
		string time:units = "seconds since 2000-01-01 00:00:00" ;
	double lat(lat) ;
		string lat:units = "degrees_north" ;
		string lat:long_name = "latitude" ;
		string lat:standard_name = "latitude" ;
		lat:valid_min = 53.000462962963 ;
		lat:valid_max = 65.999537037037 ;
	double lon(lon) ;
		string lon:units = "degrees_east" ;
		string lon:long_name = "longitude" ;
		string lon:standard_name = "longitude" ;
```

```
            lon:valid_min = 8.99982529699511 ;
            lon:valid_max = 31.0001747030049 ;

// global attributes:
            string :TileSize = "877:1049" ;
            string :id = "20250511_P1D_CMEMS_HROC_L3_tur-spm-chl_BAL__100m-v01.9.nc" ;
            string :title = "CMEMS HR-OC Baltic Sea transparency (spm, tur) and geophysical (chl) daily
observations mosaic" ;
            string :region = "BAL" ;
            string :tile = "" ;
            string :cmems_product_id = "OCEANCOLOUR_BAL_BGC_HR_L3_NRT_009_202" ;
            string :cmems_dataset = "cmems_obs_oc_bal_bgc_tur-spm-chl_nrt_l3-hr-mosaic_P1D-m" ;
            string :institution = "Brockmann Consult GmbH, RBINS, VITO for CMEMS, Mercator Ocean" ;
            string :source = "Sentinel-2 MSI L1C" ;
            string :history = "Calvalus 2.21 mosaic generation; Acolite 20210203; C2RCC 8.0cv; Idepix
8.0cv; HROC L2W 20201223; Calvalus 2.21 aggregation" ;
            string :references = "https://marine.copernicus.eu/" ;
            string :license = "CMEMS data license" ;
            string :tracking_id = "106f64dd-2d17-45bc-8cb1-e39324bec9ef" ;
            string :product_version = "v01.9" ;
            string :summary = "The HR-OC product has been processed from Sentinel-2 MSI L1C by
pixel identification, atmospheric correction, in-water retrieval, spatio-temporal aggregation, and tiling. It
covers six European coastal regions." ;
            string :keywords = "ocean color, water quality, reflectance, Copernicus CMEMS" ;
            string :keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science
keywords" ;
            string :Conventions = "CF-1.7" ;
            string :naming_authority = "org.esa-cci" ;
            string :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata
Convention" ;
            string :date_created = "20250511T180555Z" ;
            string :creator_name = "BC, RBINS, and VITO for CMEMS" ;
            string :creator_url = "https://www.brockmann-consult.de/" ;
            string :creator_email = "hr-oc-servicedesk@brockmann-consult.de" ;
            string :contact = "https://marine.copernicus.eu/" ;
            string :project = "CMEMS High Resolution Ocean Colour" ;
            string :cmd_data_type = "Grid" ;
            string :spatial_resolution = "100m" ;
            string :geospatial_lon_min = "8.998951781970646" ;
            string :geospatial_lon_max = "31.001048218029347" ;
            string :geospatial_lon_resolution = "0.0017470300489168414" ;
            string :geospatial_lon_units = "degrees_east" ;
            string :geospatial_lat_min = "53.0" ;
            string :geospatial_lat_max = "66.0" ;
            string :geospatial_lat_resolution = "9.25925925925926E-4" ;
            string :geospatial_lat_units = "degrees_north" ;
            string :geospatial_vertical_min = "0" ;
            string :geospatial_vertical_max = "0" ;
            string :time_coverage_start = "20250511T000000Z" ;
            string :time_coverage_stop = "20250511T235959Z" ;
            string :time_coverage_duration = "P1D" ;
            string :time_coverage_resolution = "P1D" ;
            string :platform = "Sentinel-2" ;
            string :sensor = "MSI" ;
            string :start_date = "11-MAY-2025 00:00:00.000000" ;
            string :stop_date = "11-MAY-2025 00:00:00.000000" ;
```

```
            :_NCProperties = "version=2,netcdf=4.7.3,hdf5=1.8.14," ;
```

For dataset cmems_obs_oc_<region>_bgc_optics_nrt_l3-hr-mosaic_P1D-m
(example of BAL region)

```
netcdf \20250228_P1D_CMEMS_HROC_L4-tur-spm-chl_BAL__100m-v01.2 {
dimensions:
	time = 1 ;
	lat = 14040 ;
	lon = 12594 ;
variables:
	float SPM(time, lat, lon) ;
		string SPM:long_name = "Suspended Particulate Matter derived from MSI L2R using HR-OC
L2W processor" ;
		string SPM:standard_name = "mass_concentration_of_suspended_matter_in_sea_water" ;
		string SPM:units = "g m-3" ;
		SPM:_FillValue = NaNf ;
		SPM:valid_min = 0.f ;
		string SPM:coordinates = "lat lon" ;
	float TUR(time, lat, lon) ;
		string TUR:long_name = "Turbidity derived from MSI L2R using HR-OC L2W processor" ;
		string TUR:standard_name = "sea_water_turbidity" ;
		string TUR:units = "FNU" ;
		TUR:_FillValue = NaNf ;
		TUR:valid_min = 0.f ;
		string TUR:coordinates = "lat lon" ;
	float CHL(time, lat, lon) ;
		string CHL:long_name = "Chlorophyll-a concentration derived from MSI L2R using HR-OC
L2W processor" ;
		string CHL:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
		string CHL:units = "mg m-3" ;
		CHL:_FillValue = NaNf ;
		CHL:valid_min = 0.f ;
		string CHL:coordinates = "lat lon" ;
	int crs ;
		string crs:wkt = "GEOGCS[\"WGS 84\", \n  DATUM[\"World Geodetic System 1984\", \n
SPHEROID[\"WGS 84\", 6378137.0, 298.257223563, AUTHORITY[\"EPSG\",\"7030\"]], \n
AUTHORITY[\"EPSG\",\"6326\"]], \n  PRIMEM[\"Greenwich\", 0.0, AUTHORITY[\"EPSG\",\"8901\"]], \n
UNIT[\"degree\", 0.017453292519943295], \n  AXIS[\"Geodetic longitude\", EAST], \n  AXIS[\"Geodetic
latitude\", NORTH], \n  AUTHORITY[\"EPSG\",\"4326\"]]" ;
		string crs:i2m = "0.0017470300489168414,0.0,0.0,-9.25925925925926E-
4,8.998951781970646,66.0" ;
	double time(time) ;
		string time:standard_name = "time" ;
		string time:axis = "T" ;
		string time:calendar = "Gregorian" ;
		string time:units = "seconds since 2000-01-01 00:00:00" ;
	double lat(lat) ;
		string lat:units = "degrees_north" ;
		string lat:long_name = "latitude" ;
		string lat:standard_name = "latitude" ;
		lat:valid_min = 53.000462962963 ;
		lat:valid_max = 65.999537037037 ;
	double lon(lon) ;
		string lon:units = "degrees_east" ;
		string lon:long_name = "longitude" ;
		string lon:standard_name = "longitude" ;
		lon:valid_min = 8.99982529699511 ;
		lon:valid_max = 31.0001747030049 ;
```

```
// global attributes:
		string :TileSize = "877:1049" ;
		string :id = "20250228_P1D_CMEMS_HROC_L4_tur-spm-chl_BAL__100m-v01.2.nc" ;
		string :title = "CMEMS HR-OC Baltic Sea transparency (spm, tur) and geophysical (chl) daily
interpolated observations mosaic" ;
		string :region = "BAL" ;
		string :tile = "" ;
		string :cmems_product_id = "OCEANCOLOUR_BAL_BGC_HR_L4_NRT_009_208" ;
		string :cmems_dataset = "cmems_obs_oc_bal_bgc_tur-spm-chl_nrt_l4-hr-mosaic_P1D-m" ;
		string :institution = "Brockmann Consult GmbH, RBINS, VITO for CMEMS, Mercator Ocean" ;
		string :source = "Sentinel-2 MSI L1C" ;
		string :history = "Calvalus 2.21 mosaic generation; Acolite 20210203; C2RCC 8.0cv; Idepix
8.0cv; HROC L2W 20201223; Calvalus 2.21 aggregation" ;
		string :references = "https://marine.copernicus.eu/" ;
		string :license = "CMEMS data license" ;
		string :tracking_id = "f46e067c-5fdc-4f84-bb70-13155d61666e" ;
		string :product_version = "v01.2" ;
		string :summary = "The HR-OC product has been processed from Sentinel-2 MSI L1C by
pixel identification, atmospheric correction, in-water retrieval, spatio-temporal aggregation, and tiling. It
covers six European coastal regions." ;
		string :keywords = "ocean color, water quality, reflectance, Copernicus CMEMS" ;
		string :keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science
keywords" ;
		string :Conventions = "CF-1.7" ;
		string :naming_authority = "org.esa-cci" ;
		string :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata
Convention" ;
		string :date_created = "20250318T142945Z" ;
		string :creator_name = "BC, RBINS, and VITO for CMEMS" ;
		string :creator_url = "https://www.brockmann-consult.de/" ;
		string :creator_email = "hr-oc-servicedesk@brockmann-consult.de" ;
		string :contact = "https://marine.copernicus.eu/" ;
		string :project = "CMEMS High Resolution Ocean Colour" ;
		string :cmd_data_type = "Grid" ;
		string :spatial_resolution = "100m" ;
		string :geospatial_lon_min = "8.998951781970646" ;
		string :geospatial_lon_max = "31.001048218029347" ;
		string :geospatial_lon_resolution = "0.0017470300489168414" ;
		string :geospatial_lon_units = "degrees_east" ;
		string :geospatial_lat_min = "53.0" ;
		string :geospatial_lat_max = "66.0" ;
		string :geospatial_lat_resolution = "9.25925925925926E-4" ;
		string :geospatial_lat_units = "degrees_north" ;
		string :geospatial_vertical_min = "0" ;
		string :geospatial_vertical_max = "0" ;
		string :time_coverage_start = "20250228T000000Z" ;
		string :time_coverage_stop = "20250228T235959Z" ;
		string :time_coverage_duration = "P1D" ;
		string :time_coverage_resolution = "P1D" ;
		string :platform = "Sentinel-2" ;
		string :sensor = "MSI" ;
		string :start_date = "28-FEB-2025 00:00:00.000000" ;
		string :stop_date = "28-FEB-2025 00:00:00.000000" ;

		:_NCProperties = "version=2,netcdf=4.7.3,hdf5=1.8.14," ;
```

For dataset cmems_obs_oc_<region>_bgc_optics_nrt_l3-hr-mosaic_P1D-m
(example of BAL region)

```
netcdf \20250401_P1M_CMEMS_HROC_L4-tur-spm-chl_BAL__100m-v01.2 {
dimensions:
	time = 1 ;
	lat = 14040 ;
	lon = 12594 ;
variables:
	float SPM(time, lat, lon) ;
		string SPM:long_name = "Suspended Particulate Matter derived from MSI L2R using HR-OC
L2W processor" ;
		string SPM:standard_name = "mass_concentration_of_suspended_matter_in_sea_water" ;
		string SPM:units = "g m-3" ;
		SPM:_FillValue = NaNf ;
		SPM:valid_min = 0.f ;
		string SPM:coordinates = "lat lon crs" ;
	float TUR(time, lat, lon) ;
		string TUR:long_name = "Turbidity derived from MSI L2R using HR-OC L2W processor" ;
		string TUR:standard_name = "sea_water_turbidity" ;
		string TUR:units = "FNU" ;
		TUR:_FillValue = NaNf ;
		TUR:valid_min = 0.f ;
		string TUR:coordinates = "lat lon crs" ;
	float CHL(time, lat, lon) ;
		string CHL:long_name = "Chlorophyll-a concentration derived from MSI L2R using HR-OC
L2W processor" ;
		string CHL:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
		string CHL:units = "mg m-3" ;
		CHL:_FillValue = NaNf ;
		CHL:valid_min = 0.f ;
		string CHL:coordinates = "lat lon crs" ;
	int crs ;
		string crs:crs_wkt = "GEOGCS[\"WGS 84\", \n  DATUM[\"World Geodetic System 1984\", \n
SPHEROID[\"WGS 84\", 6378137.0, 298.257223563, AUTHORITY[\"EPSG\",\"7030\"]], \n
AUTHORITY[\"EPSG\",\"6326\"]], \n  PRIMEM[\"Greenwich\", 0.0, AUTHORITY[\"EPSG\",\"8901\"]], \n
UNIT[\"degree\", 0.017453292519943295], \n  AXIS[\"Geodetic longitude\", EAST], \n  AXIS[\"Geodetic
latitude\", NORTH], \n  AUTHORITY[\"EPSG\",\"4326\"]]" ;
		string crs:wkt = "GEOGCS[\"WGS 84\", \n  DATUM[\"World Geodetic System 1984\", \n
SPHEROID[\"WGS 84\", 6378137.0, 298.257223563, AUTHORITY[\"EPSG\",\"7030\"]], \n
AUTHORITY[\"EPSG\",\"6326\"]], \n  PRIMEM[\"Greenwich\", 0.0, AUTHORITY[\"EPSG\",\"8901\"]], \n
UNIT[\"degree\", 0.017453292519943295], \n  AXIS[\"Geodetic longitude\", EAST], \n  AXIS[\"Geodetic
latitude\", NORTH], \n  AUTHORITY[\"EPSG\",\"4326\"]]" ;
		string crs:i2m = "0.0017470300489168414,0.0,0.0,-9.25925925925926E-
4,8.998951781970646,66.0" ;
		string crs:GeoTransform = "8.998951781970646,0.0017470300489168414,0.0,66.0,0.0,-
9.25925925925926E-4" ;
	double time(time) ;
		string time:standard_name = "time" ;
		string time:axis = "T" ;
		string time:calendar = "Gregorian" ;
		string time:units = "seconds since 2000-01-01 00:00:00" ;
	double lat(lat) ;
		string lat:units = "degrees_north" ;
		string lat:long_name = "latitude" ;
		string lat:standard_name = "latitude" ;
		lat:valid_min = 53.000462962963 ;
		lat:valid_max = 65.999537037037 ;
	double lon(lon) ;
		string lon:units = "degrees_east" ;
		string lon:long_name = "longitude" ;
		string lon:standard_name = "longitude" ;
```

```
            lon:valid_min = 8.99982529699511 ;
            lon:valid_max = 31.0001747030049 ;

// global attributes:
            string :TileSize = "877:1049" ;
            string :id = "20250401_P1M_CMEMS_HROC_L4_tur-spm-chl_BAL__100m-v01.2.nc" ;
            string :title = "CMEMS HR-OC Baltic Sea transparency (spm, tur) and geophysical (chl)
monthly averaged observations mosaic" ;
            string :region = "BAL" ;
            string :tile = "" ;
            string :cmems_product_id = "OCEANCOLOUR_BAL_BGC_HR_L4_NRT_009_208" ;
            string :cmems_dataset = "cmems_obs_oc_bal_bgc_tur-spm-chl_nrt_l4-hr-mosaic_P1M-m" ;
            string :institution = "Brockmann Consult GmbH, RBINS, VITO for CMEMS, Mercator Ocean" ;
            string :source = "Sentinel-2 MSI L1C" ;
            string :history = "Calvalus 2.21 mosaic generation; Acolite 20210203; C2RCC 8.0cv; Idepix
8.0cv; HROC L2W 20201223; Calvalus 2.21 aggregation" ;
            string :references = "https://marine.copernicus.eu/" ;
            string :license = "CMEMS data license" ;
            string :tracking_id = "89421ba9-595c-49f8-b403-36cf51995c21" ;
            string :product_version = "v01.2" ;
            string :summary = "The HR-OC product has been processed from Sentinel-2 MSI L1C by
pixel identification, atmospheric correction, in-water retrieval, spatio-temporal aggregation, and tiling. It
covers six European coastal regions." ;
            string :keywords = "ocean color, water quality, reflectance, Copernicus CMEMS" ;
            string :keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science
keywords" ;
            string :Conventions = "CF-1.7" ;
            string :naming_authority = "org.esa-cci" ;
            string :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata
Convention" ;
            string :date_created = "20250506T141953Z" ;
            string :creator_name = "BC, RBINS, and VITO for CMEMS" ;
            string :creator_url = "https://www.brockmann-consult.de/" ;
            string :creator_email = "hr-oc-servicedesk@brockmann-consult.de" ;
            string :contact = "https://marine.copernicus.eu/" ;
            string :project = "CMEMS High Resolution Ocean Colour" ;
            string :cmd_data_type = "Grid" ;
            string :spatial_resolution = "100m" ;
            string :geospatial_lon_min = "8.998951781970646" ;
            string :geospatial_lon_max = "31.001048218029347" ;
            string :geospatial_lon_resolution = "0.0017470300489168414" ;
            string :geospatial_lon_units = "degrees_east" ;
            string :geospatial_lat_min = "53.0" ;
            string :geospatial_lat_max = "66.0" ;
            string :geospatial_lat_resolution = "9.25925925925926E-4" ;
            string :geospatial_lat_units = "degrees_north" ;
            string :geospatial_vertical_min = "0" ;
            string :geospatial_vertical_max = "0" ;
            string :time_coverage_start = "20250401T000000Z" ;
            string :time_coverage_stop = "20250401T235959Z" ;
            string :time_coverage_duration = "P1M" ;
            string :time_coverage_resolution = "P1M" ;
            string :platform = "Sentinel-2" ;
            string :sensor = "MSI" ;
            string :start_date = "01-APR-2025 00:00:00.000000" ;
            string :stop_date = "01-APR-2025 00:00:00.000000" ;
            :_NCProperties = "version=2,netcdf=4.7.3,hdf5=1.8.14," ;
```

For dataset cmems_obs_oc_ARC_bgc_optics_nrt_l3-hr-mosaic_P1D-m
(example of ARC region – for one tile)

```
netcdf \20250511_P1D_CMEMS_HROC_L3-optics_ARC_h04v05_100m-v01.9 {
dimensions:
	time = 1 ;
	y = 8000 ;
	x = 6000 ;
variables:
	float RRS443(time, y, x) ;
		string RRS443:long_name = "Rrs 443nm from MSI L2R using C2RCC-ACOLITE/iCOR
processor" ;
		string RRS443:standard_name =
"surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
" ;
		string RRS443:units = "sr-1" ;
		RRS443:radiation_wavelength = 443.f ;
		RRS443:_FillValue = NaNf ;
		RRS443:valid_min = 0.f ;
		RRS443:valid_max = 1.f ;
		string RRS443:coordinates = "lat lon" ;
	float RRS492(time, y, x) ;
		string RRS492:long_name = "Rrs 492nm from MSI L2R using C2RCC-ACOLITE/iCOR
processor" ;
		string RRS492:standard_name =
"surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
" ;
		string RRS492:units = "sr-1" ;
		RRS492:radiation_wavelength = 492.f ;
		RRS492:_FillValue = NaNf ;
		RRS492:valid_min = 0.f ;
		RRS492:valid_max = 1.f ;
		string RRS492:coordinates = "lat lon" ;
	float RRS560(time, y, x) ;
		string RRS560:long_name = "Rrs 560nm from MSI L2R using C2RCC-ACOLITE/iCOR
processor" ;
		string RRS560:standard_name =
"surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
" ;
		string RRS560:units = "sr-1" ;
		RRS560:radiation_wavelength = 560.f ;
		RRS560:_FillValue = NaNf ;
		RRS560:valid_min = 0.f ;
		RRS560:valid_max = 1.f ;
		string RRS560:coordinates = "lat lon" ;
	float RRS665(time, y, x) ;
		string RRS665:long_name = "Rrs 665nm from MSI L2R using C2RCC-ACOLITE/iCOR
processor" ;
		string RRS665:standard_name =
"surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
" ;
		string RRS665:units = "sr-1" ;
		RRS665:radiation_wavelength = 665.f ;
		RRS665:_FillValue = NaNf ;
		RRS665:valid_min = 0.f ;
		RRS665:valid_max = 1.f ;
		string RRS665:coordinates = "lat lon" ;
	float RRS704(time, y, x) ;
```

```
            string RRS704:long_name = "Rrs 704nm from MSI L2R using C2RCC-ACOLITE/iCOR
processor" ;
            string RRS704:standard_name =
"surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
" ;
            string RRS704:units = "sr-1" ;
            RRS704:radiation_wavelength = 704.f ;
            RRS704:_FillValue = NaNf ;
            RRS704:valid_min = 0.f ;
            RRS704:valid_max = 1.f ;
            string RRS704:coordinates = "lat lon" ;
      float RRS740(time, y, x) ;
            string RRS740:long_name = "Rrs 740nm from MSI L2R using C2RCC-ACOLITE/iCOR
processor" ;
            string RRS740:standard_name =
"surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
" ;
            string RRS740:units = "sr-1" ;
            RRS740:radiation_wavelength = 740.f ;
            RRS740:_FillValue = NaNf ;
            RRS740:valid_min = 0.f ;
            RRS740:valid_max = 1.f ;
            string RRS740:coordinates = "lat lon" ;
      float RRS783(time, y, x) ;
            string RRS783:long_name = "Rrs 783nm from MSI L2R using C2RCC-ACOLITE/iCOR
processor" ;
            string RRS783:standard_name =
"surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
" ;
            string RRS783:units = "sr-1" ;
            RRS783:radiation_wavelength = 783.f ;
            RRS783:_FillValue = NaNf ;
            RRS783:valid_min = 0.f ;
            RRS783:valid_max = 1.f ;
            string RRS783:coordinates = "lat lon" ;
      float RRS865(time, y, x) ;
            string RRS865:long_name = "Rrs 865nm from MSI L2R using C2RCC-ACOLITE/iCOR
processor" ;
            string RRS865:standard_name =
"surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
" ;
            string RRS865:units = "sr-1" ;
            RRS865:radiation_wavelength = 865.f ;
            RRS865:_FillValue = NaNf ;
            RRS865:valid_min = 0.f ;
            RRS865:valid_max = 1.f ;
            string RRS865:coordinates = "lat lon" ;
      float RRS443_UNC(time, y, x) ;
            string RRS443_UNC:long_name = "Relative uncertainty of Rrs at 443nm from MSI L2R using
C2RCC-ACOLITE/iCOR processor" ;
            string RRS443_UNC:units = "1" ;
            RRS443_UNC:radiation_wavelength = 443.f ;
            RRS443_UNC:_FillValue = NaNf ;
            RRS443_UNC:valid_min = 0.f ;
            RRS443_UNC:valid_max = 1.f ;
            string RRS443_UNC:coordinates = "lat lon" ;
      float RRS492_UNC(time, y, x) ;
            string RRS492_UNC:long_name = "Relative uncertainty of Rrs at 492nm from MSI L2R using
C2RCC-ACOLITE/iCOR processor" ;
```

```
            string RRS492_UNC:units = "1" ;
            RRS492_UNC:radiation_wavelength = 492.f ;
            RRS492_UNC:_FillValue = NaNf ;
            RRS492_UNC:valid_min = 0.f ;
            RRS492_UNC:valid_max = 1.f ;
            string RRS492_UNC:coordinates = "lat lon" ;
      float RRS560_UNC(time, y, x) ;
            string RRS560_UNC:long_name = "Relative uncertainty of Rrs at 560nm from MSI L2R using
C2RCC-ACOLITE/iCOR processor" ;
            string RRS560_UNC:units = "1" ;
            RRS560_UNC:radiation_wavelength = 560.f ;
            RRS560_UNC:_FillValue = NaNf ;
            RRS560_UNC:valid_min = 0.f ;
            RRS560_UNC:valid_max = 1.f ;
            string RRS560_UNC:coordinates = "lat lon" ;
      float RRS665_UNC(time, y, x) ;
            string RRS665_UNC:long_name = "Relative uncertainty of Rrs at 665nm from MSI L2R using
C2RCC-ACOLITE/iCOR processor" ;
            string RRS665_UNC:units = "1" ;
            RRS665_UNC:radiation_wavelength = 665.f ;
            RRS665_UNC:_FillValue = NaNf ;
            RRS665_UNC:valid_min = 0.f ;
            RRS665_UNC:valid_max = 1.f ;
            string RRS665_UNC:coordinates = "lat lon" ;
      float RRS704_UNC(time, y, x) ;
            string RRS704_UNC:long_name = "Relative uncertainty of Rrs at 704nm from MSI L2R using
C2RCC-ACOLITE/iCOR processor" ;
            string RRS704_UNC:units = "1" ;
            RRS704_UNC:radiation_wavelength = 704.f ;
            RRS704_UNC:_FillValue = NaNf ;
            RRS704_UNC:valid_min = 0.f ;
            RRS704_UNC:valid_max = 1.f ;
            string RRS704_UNC:coordinates = "lat lon" ;
      float RRS740_UNC(time, y, x) ;
            string RRS740_UNC:long_name = "Relative uncertainty of Rrs at 740nm from MSI L2R using
C2RCC-ACOLITE/iCOR processor" ;
            string RRS740_UNC:units = "1" ;
            RRS740_UNC:radiation_wavelength = 740.f ;
            RRS740_UNC:_FillValue = NaNf ;
            RRS740_UNC:valid_min = 0.f ;
            RRS740_UNC:valid_max = 1.f ;
            string RRS740_UNC:coordinates = "lat lon" ;
      float RRS783_UNC(time, y, x) ;
            string RRS783_UNC:long_name = "Relative uncertainty of Rrs at 783nm from MSI L2R using
C2RCC-ACOLITE/iCOR processor" ;
            string RRS783_UNC:units = "1" ;
            RRS783_UNC:radiation_wavelength = 783.f ;
            RRS783_UNC:_FillValue = NaNf ;
            RRS783_UNC:valid_min = 0.f ;
            RRS783_UNC:valid_max = 1.f ;
            string RRS783_UNC:coordinates = "lat lon" ;
      float RRS865_UNC(time, y, x) ;
            string RRS865_UNC:long_name = "Relative uncertainty of Rrs at 865nm from MSI L2R using
C2RCC-ACOLITE/iCOR processor" ;
            string RRS865_UNC:units = "1" ;
            RRS865_UNC:radiation_wavelength = 865.f ;
            RRS865_UNC:_FillValue = NaNf ;
            RRS865_UNC:valid_min = 0.f ;
            RRS865_UNC:valid_max = 1.f ;
```

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            string RRS865_UNC:coordinates = "lat lon" ;
      float BBP443(time, y, x) ;
            string BBP443:long_name = "Backscattering at 443nm from MSI L2R using C2RCC-
ACOLITE/iCOR processor" ;
            string BBP443:standard_name =
"volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water" ;
            string BBP443:units = "m-1" ;
            BBP443:radiation_wavelength = 443.f ;
            BBP443:_FillValue = NaNf ;
            BBP443:valid_min = 0.f ;
            BBP443:valid_max = 10.f ;
            string BBP443:coordinates = "lat lon" ;
      float BBP492(time, y, x) ;
            string BBP492:long_name = "Backscattering at 492nm from MSI L2R using C2RCC-
ACOLITE/iCOR processor" ;
            string BBP492:standard_name =
"volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water" ;
            string BBP492:units = "m-1" ;
            BBP492:radiation_wavelength = 492.f ;
            BBP492:_FillValue = NaNf ;
            BBP492:valid_min = 0.f ;
            BBP492:valid_max = 10.f ;
            string BBP492:coordinates = "lat lon" ;
      float BBP560(time, y, x) ;
            string BBP560:long_name = "Backscattering at 560nm from MSI L2R using C2RCC-
ACOLITE/iCOR processor" ;
            string BBP560:standard_name =
"volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water" ;
            string BBP560:units = "m-1" ;
            BBP560:radiation_wavelength = 560.f ;
            BBP560:_FillValue = NaNf ;
            BBP560:valid_min = 0.f ;
            BBP560:valid_max = 10.f ;
            string BBP560:coordinates = "lat lon" ;
      float BBP665(time, y, x) ;
            string BBP665:long_name = "Backscattering at 665nm from MSI L2R using C2RCC-
ACOLITE/iCOR processor" ;
            string BBP665:standard_name =
"volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water" ;
            string BBP665:units = "m-1" ;
            BBP665:radiation_wavelength = 665.f ;
            BBP665:_FillValue = NaNf ;
            BBP665:valid_min = 0.f ;
            BBP665:valid_max = 10.f ;
            string BBP665:coordinates = "lat lon" ;
      float BBP704(time, y, x) ;
            string BBP704:long_name = "Backscattering at 704nm from MSI L2R using C2RCC-
ACOLITE/iCOR processor" ;
            string BBP704:standard_name =
"volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water" ;
            string BBP704:units = "m-1" ;
            BBP704:radiation_wavelength = 704.f ;
            BBP704:_FillValue = NaNf ;
            BBP704:valid_min = 0.f ;
            BBP704:valid_max = 10.f ;
            string BBP704:coordinates = "lat lon" ;
      float BBP740(time, y, x) ;
            string BBP740:long_name = "Backscattering at 740nm from MSI L2R using C2RCC-
ACOLITE/iCOR processor" ;
```

```
            string BBP740:standard_name =
"volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water" ;
            string BBP740:units = "m-1" ;
            BBP740:radiation_wavelength = 740.f ;
            BBP740:_FillValue = NaNf ;
            BBP740:valid_min = 0.f ;
            BBP740:valid_max = 10.f ;
            string BBP740:coordinates = "lat lon" ;
      float BBP783(time, y, x) ;
            string BBP783:long_name = "Backscattering at 783nm from MSI L2R using C2RCC-
ACOLITE/iCOR processor" ;
            string BBP783:standard_name =
"volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water" ;
            string BBP783:units = "m-1" ;
            BBP783:radiation_wavelength = 783.f ;
            BBP783:_FillValue = NaNf ;
            BBP783:valid_min = 0.f ;
            BBP783:valid_max = 10.f ;
            string BBP783:coordinates = "lat lon" ;
      float BBP865(time, y, x) ;
            string BBP865:long_name = "Backscattering at 865nm from MSI L2R using C2RCC-
ACOLITE/iCOR processor" ;
            string BBP865:standard_name =
"volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water" ;
            string BBP865:units = "m-1" ;
            BBP865:radiation_wavelength = 865.f ;
            BBP865:_FillValue = NaNf ;
            BBP865:valid_min = 0.f ;
            BBP865:valid_max = 10.f ;
            string BBP865:coordinates = "lat lon" ;
      int crs ;
            string crs:wkt = "PROJCS[\"unnamed\",GEOGCS[\"WGS
84\",DATUM[\"WGS_1984\",SPHEROID[\"WGS
84\",6378137,298.257223563,AUTHORITY[\"EPSG\",\"7030\"]],TOWGS84[0,0,0,0,0,0,0],AUTHORITY[\
"EPSG\",\"6326\"]],PRIMEM[\"Greenwich\",0,AUTHORITY[\"EPSG\",\"8901\"]],UNIT[\"degree\",0.017453
2925199433,AUTHORITY[\"EPSG\",\"9108\"]],AUTHORITY[\"EPSG\",\"4326\"]],PROJECTION[\"Lamber
t_Azimuthal_Equal_Area\"],PARAMETER[\"latitude_of_center\",90],PARAMETER[\"longitude_of_center
\",0],PARAMETER[\"false_easting\",0],PARAMETER[\"false_northing\",0],UNIT[\"Meter\",1],AUTHORIT
Y[\"epsg\",\"6931\"]]" ;
            string crs:i2m = "100.0,0.0,0.0,-100.0,-1200000.0,-800000.0" ;
      double time(time) ;
            string time:standard_name = "time" ;
            string time:axis = "T" ;
            string time:calendar = "Gregorian" ;
            string time:units = "seconds since 2000-01-01 00:00:00" ;
      double lat(y, x) ;
            string lat:units = "degrees_north" ;
            string lat:long_name = "latitude coordinate" ;
            string lat:standard_name = "latitude" ;
      double lon(y, x) ;
            string lon:units = "degrees_east" ;
            string lon:long_name = "longitude coordinate" ;
            string lon:standard_name = "longitude" ;
      double y(y) ;
            string y:units = "m" ;
            string y:long_name = "y coordinate of projection" ;
            string y:standard_name = "projection_y_coordinate" ;
            y:valid_min = -1600000. ;
            y:valid_max = -800000. ;
```

```
        double x(x) ;
                string x:units = "m" ;
                string x:long_name = "x coordinate of projection" ;
                string x:standard_name = "projection_x_coordinate" ;
                x:valid_min = -1200000. ;
                x:valid_max = -600000. ;

// global attributes:
                string :TileSize = "500:500" ;
                string :id = "20250511_P1D_CMEMS_HROC_L3_optics_ARC_h04v05_100m-v01.9.nc" ;
                string :title = "CMEMS HR-OC Arctic Region optical (rrs, bbp) daily observations tile" ;
                string :region = "ARC" ;
                string :tile = "h04v05" ;
                string :cmems_product_id = "OCEANCOLOUR_ARC_BGC_HR_L3_NRT_009_201" ;
                string :cmems_dataset = "cmems_obs_oc_arc_bgc_optics_nrt_l3-hr_P1D-m" ;
                string :institution = "Brockmann Consult GmbH, RBINS, VITO for CMEMS, Mercator Ocean" ;
                string :source = "Sentinel-2 MSI L1C" ;
                string :history = "Calvalus 2.21 mosaic generation; Acolite 20210203; C2RCC 8.0cv; Idepix
8.0cv; HROC L2W 20201223; Calvalus 2.21 aggregation" ;
                string :references = "https://marine.copernicus.eu/" ;
                string :license = "CMEMS data license" ;
                string :tracking_id = "e5404d73-3199-4a6d-b973-9ddc4d1a8dae" ;
                string :product_version = "v01.9" ;
                string :summary = "The HR-OC product has been processed from Sentinel-2 MSI L1C by
pixel identification, atmospheric correction, in-water retrieval, spatio-temporal aggregation, and tiling. It
covers six European coastal regions." ;
                string :keywords = "ocean color, water quality, reflectance, Copernicus CMEMS" ;
                string :keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science
keywords" ;
                string :Conventions = "CF-1.7" ;
                string :naming_authority = "org.esa-cci" ;
                string :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata
Convention" ;
                string :date_created = "20250512T024200Z" ;
                string :creator_name = "BC, RBINS, and VITO for CMEMS" ;
                string :creator_url = "https://www.brockmann-consult.de/" ;
                string :creator_email = "hr-oc-servicedesk@brockmann-consult.de" ;
                string :contact = "https://marine.copernicus.eu/" ;
                string :project = "CMEMS High Resolution Ocean Colour" ;
                string :cmd_data_type = "Grid" ;
                string :spatial_resolution = "100m" ;
                string :geospatial_y_min = "-1600000.0" ;
                string :geospatial_y_max = "-800000.0" ;
                string :geospatial_y_resolution = "100.0" ;
                string :geospatial_y_units = "m" ;
                string :geospatial_x_min = "-1200000.0" ;
                string :geospatial_x_max = "-600000.0" ;
                string :geospatial_x_resolution = "100.0" ;
                string :geospatial_x_units = "m" ;
                string :geospatial_vertical_min = "0" ;
                string :geospatial_vertical_max = "0" ;
                string :time_coverage_start = "20250511T000000Z" ;
                string :time_coverage_stop = "20250511T235959Z" ;
                string :time_coverage_duration = "P1D" ;
                string :time_coverage_resolution = "P1D" ;
                string :platform = "Sentinel-2" ;
                string :sensor = "MSI" ;
                string :auto_grouping = "RRS*UNC:RRS*:BBP*" ;
                string :start_date = "11-MAY-2025 00:00:00.000000" ;
```

```
        string :stop_date = "11-MAY-2025 00:00:00.000000" ;
        :_NCProperties = "version=2,netcdf=4.7.3,hdf5=1.8.14," ;
```